

PASSAGE ISLAND, MI, USA

WMO: 994090

Lat: 48.223N

Lon: 88.366W

Elev: 195

StdP: 99.00

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -20.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 19.7	(k) -9.0	(l) 18.3	(m) -8.3	(n) 8.7	(o) 330	(p) 0.905

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.5	(c) 20.6	(d) N/A	(e) 19.6	(f) N/A	(g) 18.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.9	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.4	(b) 15.9	(c) 14.3	(d) DB	(e) -24.8	(f) 22.9	(g) 2.1	(h) 2.1	(i) -26.3	(j) 24.4	(k) -27.5	(l) 25.6	(m) -28.7	(n) 26.7	(o) -30.2	(p) 28.2	(q) N/A
(4) 18.4	(5) 15.9	(6) 14.3	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-10.0	-8.4	-4.0	1.5	5.7	9.3	13.5	16.0	13.3	7.2	0.7	-4.4
	DBStd	9.44	6.36	5.97	5.16	3.10	2.69	2.32	3.16	2.39	3.01	3.57	4.78	5.35
	HDD10.0	2837	620	516	433	256	137	40	5	0	7	100	278	445
	HDD18.3	5444	878	749	692	506	391	271	152	79	152	344	528	703
	CDD10.0	443	0	0	0	0	4	19	114	186	107	14	0	0
	CDD18.3	9	0	0	0	0	0	0	2	6	1	0	0	0
	CDH23.3	1	0	0	0	0	0	1	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	7.0	8.3	8.1	7.2	6.7	6.2	5.4	5.4	6.0	6.6	7.8	8.3	8.0
(15) Precipitation	PrecAvg	879	47	34	42	67	88	100	96	89	104	91	68	53
	PrecMax	1050	112	84	83	196	207	204	170	137	215	167	133	95
	PrecMin	646	11	11	11	18	48	46	57	42	43	34	25	26
	PrecStd	101	19	17	16	33	31	35	29	26	37	34	26	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.7	7.8	10.1	16.3	18.4	22.0	22.0	20.7	16.3	9.9	5.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	2.3	2.1	4.4	7.5	13.5	16.2	20.2	20.8	19.1	14.5	8.5	4.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	1.1	0.7	2.8	5.8	11.1	14.5	19.0	20.0	17.9	13.0	7.7	3.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	-0.5	-0.3	1.7	4.7	9.2	12.9	18.0	19.0	16.8	11.8	6.9	2.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.0	4.8	4.0	5.1	6.7	5.9	3.5	3.4	3.3	4.2	5.1
		MCDBR	6.5	6.0	4.9	5.7	8.8	8.9	6.4	4.5	4.5	4.3	3.9	4.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.276	0.274	0.304	0.337	0.362	0.375	0.397	0.373	0.360	0.321	0.293	0.268
	taud		2.464	2.461	2.414	2.392	2.422	2.413	2.407	2.475	2.485	2.554	2.539	2.518
	Ebn at Noon		813	901	919	915	900	887	861	869	847	833	789	784
	Edh at Noon		62	78	96	109	111	113	112	100	90	72	58	53
(44) All-Sky Solar Radiation	RadAvg		1.05	2.04	3.39	4.73	5.50	5.93	5.98	5.14	3.69	2.17	1.12	0.83
	RadStd		0.12	0.22	0.31	0.49	0.37	0.41	0.33	0.38	0.25	0.27	0.12	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page